About Saeem Mehmood Sadiq

Knowledge base for the "About Saeem" RAG assistant. Every section is written to make sense on its own, because the retriever will fetch sections individually, not the whole file.

Quick Facts About Saeem

- **Full name:** Saeem Mehmood Sadiq
- **Role:** Senior Software Engineer (backend), moving into Data and AI Engineering
- **Current company:** ILI Digital
- **Experience:** 5+ years of professional software engineering, starting December 2020
- **Primary languages:** Go, Elixir, Python
- **Location:** Lahore, Pakistan
- **Email:** saimmahmood61@gmail.com
- **LinkedIn:** https://www.linkedin.com/in/saimmahmood61
- **GitHub:** https://www.github.com/saim61
- **Booking a call:** https://calendar.app.google/czDkhWsVGq97Xqa47

Professional Summary of Saeem

Saeem Mehmood Sadiq is a senior backend engineer with over five years of experience building high-performance systems in Go, Elixir, and Python. He has worked across real-time systems, iGaming, ticketing platforms, SaaS products, and distributed architectures. Saeem is known for clean engineering, pragmatic decision-making, and mentoring junior developers, and he typically owns projects end-to-end, from gathering requirements through to production reliability.

Saeem's Current Role — ILI Digital

Period: April, 2026 – Present

Saeem currently works at ILI Digital as a Senior Software Engineer, where he manages projects built in Go. Alongside this role, he is transitioning into Data and AI Engineering.

Saeem's Experience at Code Graphers (Senior Software Engineer)

Period: June 2024 – March 2026

At Code Graphers, Saeem worked as a Senior Software Engineer, mainly in Go, across software asset management and iGaming projects.

- Re-engineered the Python `imagehash` library in Go, keeping output parity across all hash types. This enabled a latency-critical image-matching pipeline to migrate to a faster Go backend.
- Built backend services in Go for a client's software asset management platform, improving reliability and observability.
- Reduced the client software's idle RAM usage from about 2 GB to about 40 MB (roughly a 98% reduction) by finding memory hotspots and leaks and optimizing allocations and data structures so nothing was retained unnecessarily while idle.

- Designed a serverless data-processing pipeline using AWS Lambda, EventBridge, DynamoDB, and CloudFormation.
- Led feature development, bug resolution, and operational oversight for 13 microservices in an iGaming ecosystem.
- Worked with senior engineers and external provider teams to architect and integrate two new live-casino provider integrations.
- Mentored junior developers on Go best practices and code review standards.

Saeem's Experience at Dycoders (Senior Software Engineer, Hybrid)

Period: July 2023 – January 2024

At Dycoders, Saeem led an Elixir/Phoenix engineering team.

- Led the team through sprints, architecture discussions, and CI/CD workflows.
- Established testing standards and code review processes, which reduced production bugs and improved deployment reliability.
- Built and optimized Phoenix and LiveView backends with a focus on maintainability and real-time performance.
- Mentored junior engineers through pair programming and structured feedback.

Saeem's Experience at Automatiq (Senior Software Engineer, Contract, Remote)

Period: May 2022 – June 2023

At Automatiq, Saeem focused on integrating external providers into a ticketing platform.

- Onboarded more than 10 new providers by integrating external systems end-to-end, covering APIs, workflows, and edge cases, which expanded platform coverage and supported business growth.
- Delivered integrations with high operational stability through resilient error handling, retries and timeouts, and clear observability for faster production debugging.
- Improved response times and platform performance by optimizing backend endpoints and database interactions so the platform handled peak usage reliably.

Saeem's Experience at InvoZone (Software Engineer, Hybrid)

Period: December 2020 – May 2022

InvoZone was where Saeem started his professional career, working on real-time gaming systems.

- Developed Elixir and Go microservices for real-time gaming systems using optimized concurrency patterns.
- Implemented gRPC and GraphQL services to improve modularity and scalability.
- Owned backend logic for multiple game modules and mentored junior engineers.

Saeem's Technical Skills

- **Backend languages:** Go, Elixir (Phoenix, OTP, LiveView), Python (FastAPI, Django)
- **Architecture:** Microservices, event-driven systems, REST, gRPC, GraphQL, WebSockets
- **Databases and data stores:** PostgreSQL, MySQL, DynamoDB, Redis

- **Data engineering:** Batch ETL pipelines, dbt, PySpark, Parquet, star-schema modeling, slowly changing dimensions (SCD2), data quality checks, idempotent and incremental loading with watermarks
- **Cloud:** AWS Lambda, EventBridge, DynamoDB, CloudFormation
- **Infrastructure and DevOps:** Docker, Docker Compose, Kubernetes, Helm, GitHub Actions, CI/CD
- **Observability:** Structured logging, Prometheus metrics, OpenTelemetry
- **Engineering practices:** Testing and TDD, code review, performance and memory optimization
- **Domains:** Real-time systems, iGaming, ticketing, SaaS, software asset management

Saeem's Personal Projects and Open Source Work

All of Saeem's personal projects are public on his GitHub: https://www.github.com/saim61

Hookline — Webhook Delivery Service (Python)

Hookline is Saeem's reliable webhook delivery service, a smaller, readable implementation of what products like Svix and Hookdeck offer. It accepts an event in milliseconds and then delivers it to customers' HTTP endpoints with retries, exponential backoff with jitter, per-endpoint circuit breakers, HMAC signatures, idempotency keys, a dead letter queue, and a permanent audit trail of every delivery attempt. The delivery worker uses a transactional outbox drained with `SELECT ... FOR UPDATE SKIP LOCKED`, so multiple workers can run concurrently without double-delivering. It includes an authenticated API, an operator dashboard, Prometheus metrics, and a Helm chart, and is covered by 292 tests at 94% coverage.

- **Stack:** Python, FastAPI, PostgreSQL, SQLAlchemy (async), Redis, Alembic, HTMX, structlog, Prometheus, OpenTelemetry, Docker, Helm, GitHub Actions
- **Link:** https://github.com/saim61/hookline

Podium — Real-Time Leaderboard System (Go)

Podium is Saeem's real-time leaderboard system written in Go. Five games run on the server as state machines, so the server computes every score itself and clients can never submit a forged score. Scores are stored durably in PostgreSQL, while Redis sorted sets act as a derived read model that can be rebuilt from Postgres at any time. Leaderboard updates reach browsers over WebSockets and travel through Redis Pub/Sub, so real-time updates keep working across multiple server instances. It also includes Argon2id authentication with refresh token rotation, period reports, rate limiting, Prometheus metrics, an OpenAPI description, and a playable demo page, and is covered by 429 tests.

- **Stack:** Go, PostgreSQL, Redis, WebSockets, sqlc, Docker, Prometheus
- **Link:** https://github.com/saim61/podium

fxflow — Foreign Exchange Rates ETL Pipeline (Python)

fxflow is Saeem's batch ETL pipeline for foreign exchange rates, built to learn data engineering fundamentals from first principles before using standard tooling. It pulls daily European Central Bank reference rates from the Frankfurter API, stores them in an immutable, append-only raw layer, transforms them into an analytics-ready fact table with dbt, validates them with data quality tests, and serves them through a FastAPI REST API. The pipeline is idempotent, uses watermarks to fetch only missing data and backfill gaps, and records every run for observability.

- **Stack:** Python, PostgreSQL, SQLAlchemy, dbt, FastAPI, Docker
- **Link:** https://github.com/saim61/fxflow

Olist Analytics Platform — E-Commerce Data Warehouse (Python)

The Olist project is Saeem's end-to-end batch data pipeline built on the public Brazilian e-commerce dataset by Olist. It turns raw CSV data into a tested star-schema warehouse and business-ready Parquet marts, answering questions such as which product categories and states drive revenue, how revenue is trending, and where deliveries are late. Data moves through source, raw, staging, and warehouse layers, with incremental extraction using watermarks, a customer dimension tracked as a slowly changing dimension (SCD2), quality gates that abort the run if checks fail, and PySpark jobs that build partitioned Parquet marts. The whole platform is reproducible with Docker.

- **Stack:** Python, PostgreSQL, PySpark, Parquet, pandas, Docker
- **Link:** https://github.com/saim61/olist

Go Message App

A real-time messaging backend built by Saeem with Go, WebSockets, Clean Architecture, JWT authentication, and PostgreSQL.

imagehash — Open-Source Contribution

Saeem fixed a bug in the ColorHash algorithm of the Python `imagehash` library. His pull request #216 was accepted.

Saeem's University Education

Saeem holds a BSc in Computer Science from FAST-NUCES (National University of Computer and Emerging Sciences), Islamabad campus.

His final year project was a Twitter Cyber Bullying Detector, which analyzed tweets from a user's Twitter home feed and alerted users about accounts that were potential bullies. During university, Saeem particularly enjoyed the Software for Mobile Devices course, where he learned to build native Android apps, as well as mathematics courses.

Saeem's A Levels

Saeem completed his A Levels in 2013 at Arab Unity School in Dubai, UAE, under the Edexcel board. His subjects were Mathematics, Physics, and Chemistry.

Saeem's O Levels

Saeem completed his O Levels in 2011 at The Westminster School in Dubai, UAE, under the Cambridge (CIE) board. His subjects were Mathematics, Physics, Chemistry, Information and Communication Technology, Economics, English, and Urdu.

Saeem's Background

Saeem was born and raised in the United Arab Emirates, where he completed both his O Levels and A Levels in Dubai. He then studied Computer Science at FAST-NUCES in Islamabad, Pakistan, and is now based in Lahore, Pakistan.

What Saeem Is Learning and Where His Career Is Heading

Alongside backend engineering, Saeem is transitioning into Data and AI Engineering using Python. He is also interested in Platform Engineering, Security Engineering, and AI Systems Engineering as future directions. His fxflow and Olist projects are part of this data engineering journey, and this "About Saeem" assistant is itself one of his learning projects, built to understand embeddings and retrieval-augmented generation (RAG) hands-on.

Saeem's Hobbies and Fun Facts

Outside of work, Saeem enjoys going out with friends and loves hiking. He is especially drawn to the mountains and enjoys a good hike that ends with a killer view. He has travelled to many places in the Northern Areas of Pakistan and has also explored the hilly areas around Baku, Azerbaijan. He also likes to hang out with his friends in his free time or watch movies/seasons

How to Contact Saeem

The best way to reach Saeem is by email at saimmahmood61@gmail.com. He can also be found on LinkedIn and GitHub, and anyone can book a meeting with him through his calendar.

- **Email:** saimmahmood61@gmail.com
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